Prototyping and Design Validation

COMP-7431: Advanced Software Engineering | Week 8
CampusConnect: validating video before scaling the build

Prototype the uncertainty. Observe the evidence. Revise the design.

Monday Morning: Week 7 Made the Design Testable

Week 7 converted architecture into observable promises. Week 8 asks whether those promises work in a real journey.

WEEK 7 CONTRACT

“The user can hear a short answer, inspect the same answer as text, see its source, and continue in text if voice fails.”

WEEK 8 VALIDATION QUESTION

Can a student complete that journey when an avatar video is added—without confusion, delay, distraction, or lost access?

Contracts tell us what should happen. Prototypes reveal what people actually understand and do.

A Prototype Makes a Question Tangible

QUESTION

Will students understand what the presenter video does—and what it does not do?

Example: Does the label make it clear that the clip is prerecorded and AI-generated?

EXPERIENCE

Build only enough of the journey for a user to react, choose, and recover.

Example: A playable clip beside a working chatbot can test placement and fallback without building live video conversation.

EVIDENCE

Observe behavior that can confirm, weaken, or reject the design assumption.

Example: The student mutes the clip, reads the transcript, and still completes the task.

If the prototype cannot change a decision, it is probably a demonstration—not a validation instrument.

Prototype Learning—Not the Entire Product

BUILD JUST ENOUGH

- One representative user journey
- The risky interaction or dependency
- Visible success, error, and fallback states

Example: Generate one 30–60 second presenter clip and place it beside the existing chatbot.

DEFER WHAT DOES NOT ANSWER THE QUESTION

- Production-scale data and infrastructure
- Every screen, persona, and edge case
- Perfect branding, animation, or performance

Example: Do not build a live avatar service merely to learn whether students want an optional introduction.

Low cost is a design constraint: it keeps the team willing to discard a weak idea.

Fidelity Has Several Dimensions

VISUAL

How finished does it look?

EXAMPLE
Sketch, wireframe, or branded page.

INTERACTION

How much of the journey behaves?

EXAMPLE
Clickable controls, real captions, working text fallback.

CONTENT + DATA

How realistic are scripts, answers, citations, and errors?

EXAMPLE
Real CampusConnect wording with simulated backend responses.

A prototype may look polished but still simulate the backend—or look rough while testing a real integration.

Choose the Cheapest Fidelity That Can Answer the Question

1 STORYBOARD	**Test sequence and concept**	*Can users explain where the video belongs?*
2 CLICKABLE MOCKUP	**Test navigation and labels**	*Can users find transcript, mute, and text fallback?*
3 CONCIERGE / WIZARD-OF-OZ	**Simulate intelligence behind the screen**	*Does the interaction make sense before APIs exist?*
4 FUNCTIONAL SLICE	**Test the risky integration**	*Does the real video load, play, fail, and recover correctly?*

Do not pay for functional fidelity when a sketch can answer the question—but do not use a sketch to judge latency or media playback.

Prototype the Riskiest Assumption First

VALUE

Does the avatar help students begin—or merely decorate the page?

Test task start and comprehension before polishing.

USABILITY

Can users control, mute, replay, and ignore the video?

Observe the journey without coaching.

FEASIBILITY

Will the embed, captions, and fallback work in the target browser?

Test the real media path early.

TRUST + ETHICS

Will users mistake a synthetic presenter for a live person?

Test disclosure and consent language.

Risk × uncertainty determines what to prototype first—not which feature looks most impressive.

CampusConnect Receives a New—but Ambiguous—Request

NEW END-USER EVIDENCE

“The voice chatbot is useful, but I would feel more comfortable if someone briefly showed me what it can do.”

- Some students want a visible presenter before they begin.
- Others find talking heads distracting or prefer silent use.
- Accessibility staff require captions, transcript, keyboard control, and a complete non-video path.
- IT Support does not want a prerecorded presenter to appear live or authoritative beyond approved guidance.

The request is not “add an avatar.” The design problem is to improve orientation without reducing truth, access, or control.

Translate the Request into Testable Hypotheses

HYPOTHESIS 1 — ORIENTATION

A short presenter clip helps students understand what CampusConnect can and cannot do.

Evidence: After viewing, the student accurately names the chatbot's purpose and limits.

HYPOTHESIS 2 — ACCESS

Students can obtain the same meaning without hearing the video.

Evidence: Muted users complete the introduction using captions or transcript.

HYPOTHESIS 3 — TRUST

Clear disclosure prevents the presenter from being mistaken for a live human.

Evidence: The student describes it as an AI-generated, prerecorded introduction.

A hypothesis is useful only when the team can name the behavior that would support or contradict it.

Test a Complete Student Journey

1
Open the page
→
2
Notice AI disclosure
→
3
Play—or skip—the video
→
4
Use captions or transcript
→
5
Ask the chatbot
→
6
Continue in voice or text

TASK PROMPT

"Find out how CampusConnect can help with a password-reset question. You may use the page in any way you prefer."

Do not say: "Click Play, turn on captions, then ask the bot." That instruction would hide usability problems.

A realistic task gives the user a goal; the design must reveal the path.

The Video Experience Needs Explicit States

READY
Poster, disclosure, Play, transcript link

→

PLAYING
Pause, mute, captions, progress

→

ENDED
Replay and clear path to chatbot

→

BLOCKED / FAILED
Transcript and open-video fallback

→

SKIPPED
Chat remains fully usable

WHY STATES MATTER

A prototype that shows only a successful playback cannot validate recovery.

Example: If the browser blocks autoplay, the page must still show a visible Play button and transcript—not an empty rectangle.

Every state needs a visible user action, a recoverable next step, and a test.

Write a Prototype Contract Before Building

REAL IN THIS PROTOTYPE

- Real 30–60 second HeyGen video
- Real disclosure, captions or transcript
- Real Play, pause, mute, replay, and skip behavior
- Real link from video area to the chatbot

SIMULATED OR DEFERRED

- A small set of known chatbot answers may be staged
- Analytics may be recorded manually
- The presenter does not answer live questions
- Production scaling and personalization are out of scope

Label simulations honestly. A prototype may simplify the system; it may not misrepresent what is real.

An Avatar Must Have a Job

ORIENT

Explain the purpose, limitations, and next action.

Example: "Ask a campus IT question; CampusConnect answers from approved support guides."

REASSURE

Reduce uncertainty without claiming human authority or live presence.

Example: Use a calm introduction—not "I am your personal IT representative."

HAND OFF

Move attention to the working product quickly.

Example: End with "You can skip this video and type your question below."

If the video has no distinct job, remove it. More media is not automatically a better experience.

Stock Avatar or Personal Digital Twin?

Decision factor	Stock avatar	Personal digital twin
45-minute lab fit	Fastest path; choose and script	Setup may consume the lab unless prepared
Identity + consent	No student likeness is created	Requires the represented person’s explicit consent
Presence	Neutral and consistent	More personal and recognizable
Stewardship	Manage script, disclosure, and asset use	Also govern likeness, access, reuse, and removal

Recommended default: use a stock avatar. Choose a digital twin only when consent and setup are already complete.

A 30–60 Second Script Can Carry the Entire Contract

SCRIPT EXAMPLE

“Hello. I’m an AI-generated presenter introducing CampusConnect. It answers campus IT questions from approved support guides and shows the source used. It cannot view your account or change your password. You may play this video with captions, read the transcript, or skip it entirely. To begin, type or speak a question below. If CampusConnect lacks evidence, it will say so and direct you to IT Support.”

DISCLOSE

AI-generated and prerecorded

BOUND

Purpose and limits

ACCESS

Captions, transcript, and skip

ACTION

Clear next step

The script is not marketing copy. It is a compact user-facing contract.

Disclosure Must Appear Before the User Interprets the Presenter

IDENTITY

State that the presenter is AI-generated.

Visible example: “AI-generated presenter • prerecorded introduction.”

ROLE

State what the clip does—and what answers questions.

Example: “This video introduces the chatbot; CampusConnect generates the answers.”

CONSENT + CONTROL

Use only a person’s digital twin with their permission and preserve their ability to govern the likeness.

Example: Do not create or reuse a colleague’s twin because their photo or recording is available.

Disclosure hidden after playback is too late; the user has already formed an interpretation.

Design for Mute-Safe Use from the Start

THE VIDEO PLAYER

- Starts only after a clear user action—or remains muted
- Offers pause, replay, mute, and caption controls
- Never blocks access to the chatbot

Example: If audible autoplay is blocked, show a Play button rather than reporting a system failure.

THE TEXT EQUIVALENT

- Provides synchronized captions for spoken content
- Places a readable transcript beside or directly below the player
- Carries the same meaning and next action

Example: A student in a quiet library can understand the introduction without sound.

Mute-safe means the purpose and action survive when audio never plays.

Accessibility Is Observable Behavior

PERCEIVABLE

Captions and transcript carry all spoken meaning.

EXAMPLE
Do not encode a critical instruction only in a facial gesture.

OPERABLE

Keyboard users can reach Play, pause, captions, transcript, and chat.

EXAMPLE
Visible focus shows which control Enter will activate.

UNDERSTANDABLE

Labels explain what the video is and what happens next.

EXAMPLE
Use “Play AI introduction,” not an unlabeled triangle icon.

An accessibility claim becomes useful only when a tester can observe and repeat it.

Prototype Failure and Recovery—Not Just Playback

Condition	Prototype behavior	Preserved outcome
VIDEO BLOCKED	Show poster, disclosure, transcript, and a direct Play/open link.	The chatbot remains usable.
CAPTIONS MISSING	Do not treat auto-generated text as verified; provide the reviewed transcript.	The same message remains available.
EMBED FAILS	Replace the player with a plain explanation and hosted-video link.	No empty frame or endless spinner.
USER SKIPS	Move focus to the chatbot and preserve the complete path.	Skipping is a valid choice—not an error.

Graceful degradation removes convenience, not meaning, truth, or task completion.

Observe Behavior Before Asking for Opinions

DURING THE TASK

- Give the user a realistic goal.
- Stay quiet while the user chooses a path.
- Record actions, hesitation, errors, and recovery.

AFTER THE TASK

- Ask what the student believes the presenter is.
- Ask them to repeat the chatbot’s limits.
- Ask what they would do if the video failed.

Weak question: “Do you like the avatar?” Better task: “Learn what CampusConnect can do and begin a password-reset question.”

Preference is useful context; observed task behavior is stronger design evidence.

Collect a Small, Decision-Ready Evidence Set

EFFECTIVENESS

Did the student understand the purpose and complete the next action?

EFFICIENCY

Where did the student hesitate, replay, search, or abandon?

ACCESS

Could the same task be completed muted and by keyboard?

TRUST

Did the student recognize the presenter as AI-generated and prerecorded?

RECOVERY

Could the student continue when the video was skipped or failed?

EXAMPLE OBSERVATION

“Participant found Play immediately, but searched below the chatbot for the transcript. Move the transcript link next to the player before implementation.”

Record what happened, where it happened, and what decision it informs.

Compare Evidence with Requirements

Requirement	Prototype evidence	Decision
CC-UX-08: Video is optional	Student skipped it and completed the chatbot task	PASS — preserve skip path
CC-A11Y-03: Meaning survives mute	Captions worked; transcript link was hard to find	REVISE — move transcript beside Play
CC-TRUST-02: Synthetic media is disclosed	Student assumed a live staff member until playback ended	FAIL — disclose before Play
CC-REC-04: Embed failure has fallback	Broken embed showed an empty frame	FAIL — add transcript and direct link

Validation does not ask whether the prototype is impressive. It asks whether evidence supports each requirement.

Revise the Design—and Preserve the Reason

KEEP	**Optional video and complete skip path**	*Evidence showed students could continue without watching.*
CHANGE	**Place transcript beside Play and keep it openable at all times**	*Muted users searched for it.*
REMOVE	**Audible autoplay**	*It surprised users and may be blocked by browsers.*
ADD	**Pre-play AI disclosure and an embed-failure fallback**	*Users misread the role; failure state was empty.*

GitHub trail: update the requirement or design note → commit the prototype change → open a pull request → ask a reviewer whether the evidence supports the revision.

The artifact is not merely the new screen. It is the evidence-backed decision that explains why the screen changed.

The Week 8 Lab Uses Three Tools with Distinct Jobs

HEYGEN

Create one prerecorded 30–60 second stock-avatar or consented digital-twin video.

Current free plan: up to one-minute videos; quota may be 1–3 by region.

LOVABLE

Place the video beside the chatbot and implement disclosure, controls, transcript, and fallback.

The prototype contract stays portable if the interface builder changes.

GITHUB

Version the script, design decision, evidence, and implementation change.

A draft pull request makes the revision reviewable before Week 9.

The lab goal is not "use three AI tools." It is to produce one small, testable, evidence-backed design revision.

Week 8 Ends with Evidence—Week 9 Begins with a Build

"Do not scale an assumption merely because the prototype looks finished."

- Choose fidelity from the question
- Prototype the highest risk
- Test the complete user journey
- Include access and failure states

- Compare evidence with requirements
- Keep, change, remove, or add
- Preserve the decision in GitHub
- Carry only validated behavior forward

NEXT MONDAY — WEEK 9
Implementation, Version Control, and Code Review

Validated behavior becomes issues, branches, code, tests, commits, and a reviewable pull request.